

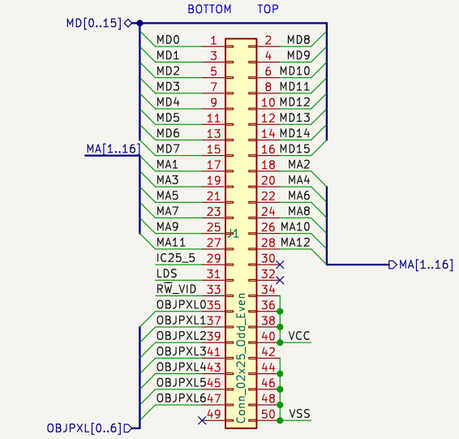


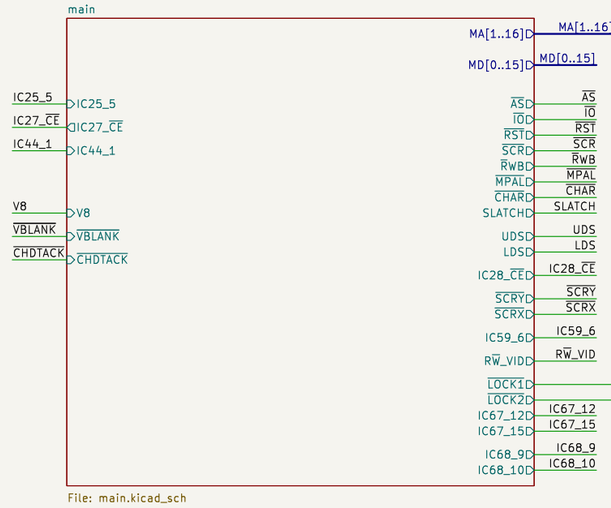
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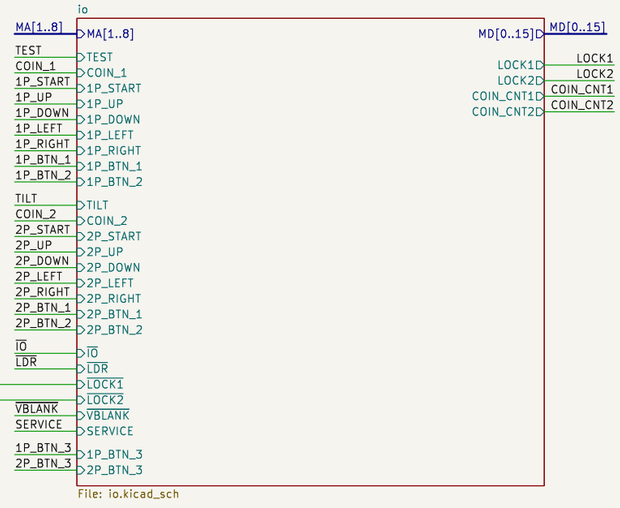
Sheet: /
File: wwfsstar.kicad_sch

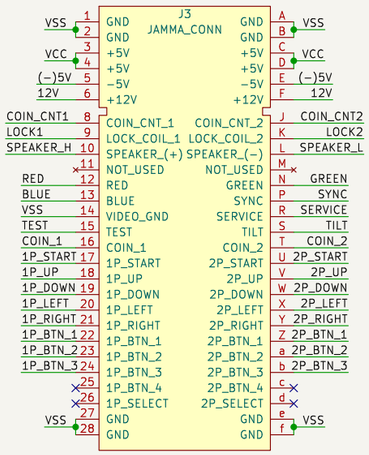
Title: WWF Super Stars

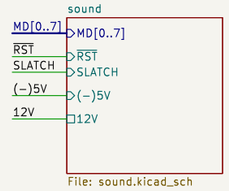
Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/20

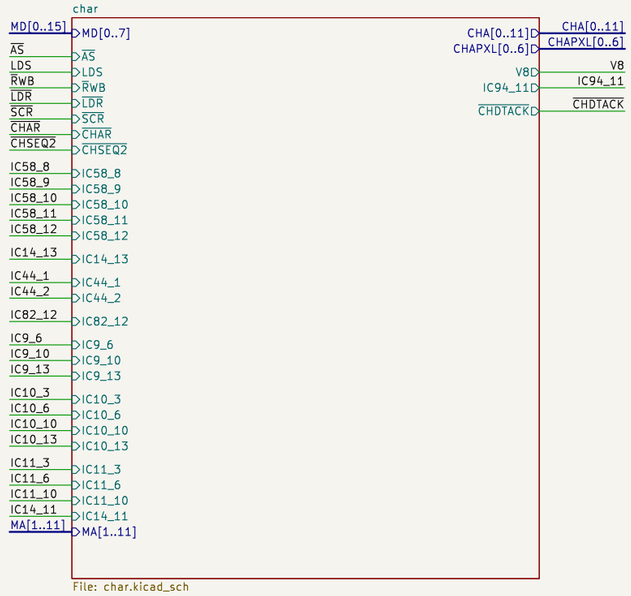


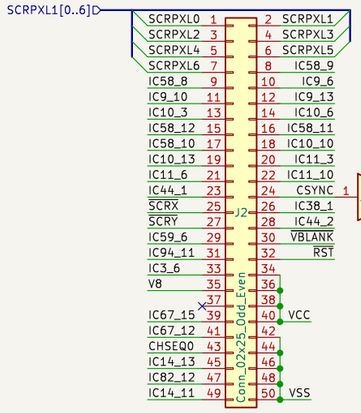


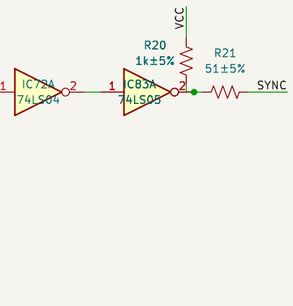


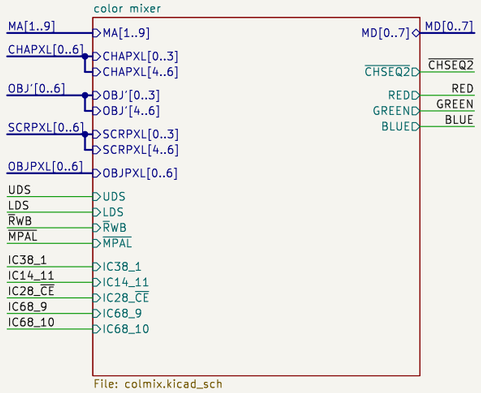












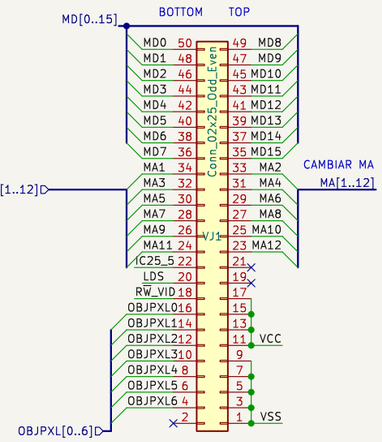
JOTEGO

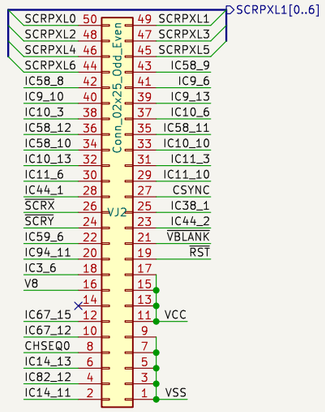
Sheet: /top/

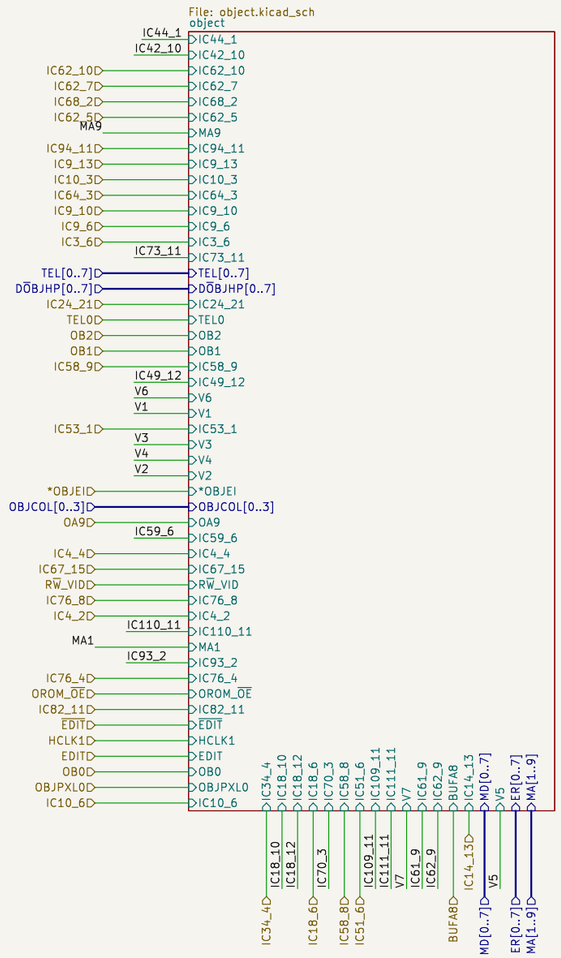
File: top.kicad_sch

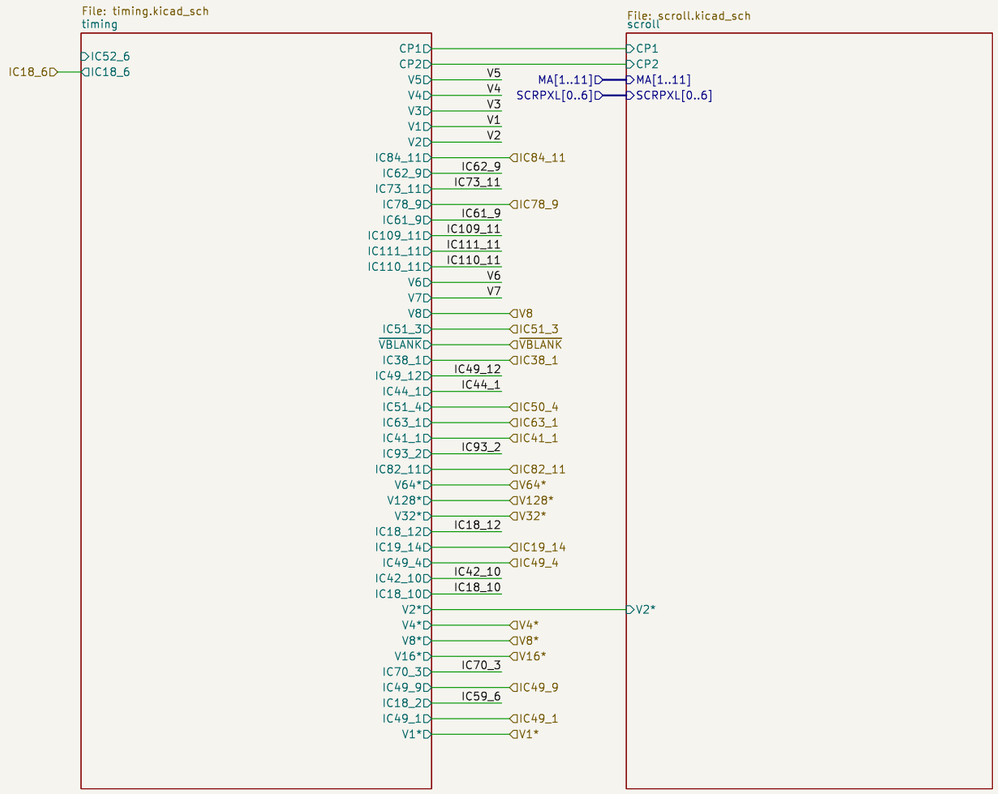
Title: Connectors and subsystems

Size: A3	Date:	**Rev:**
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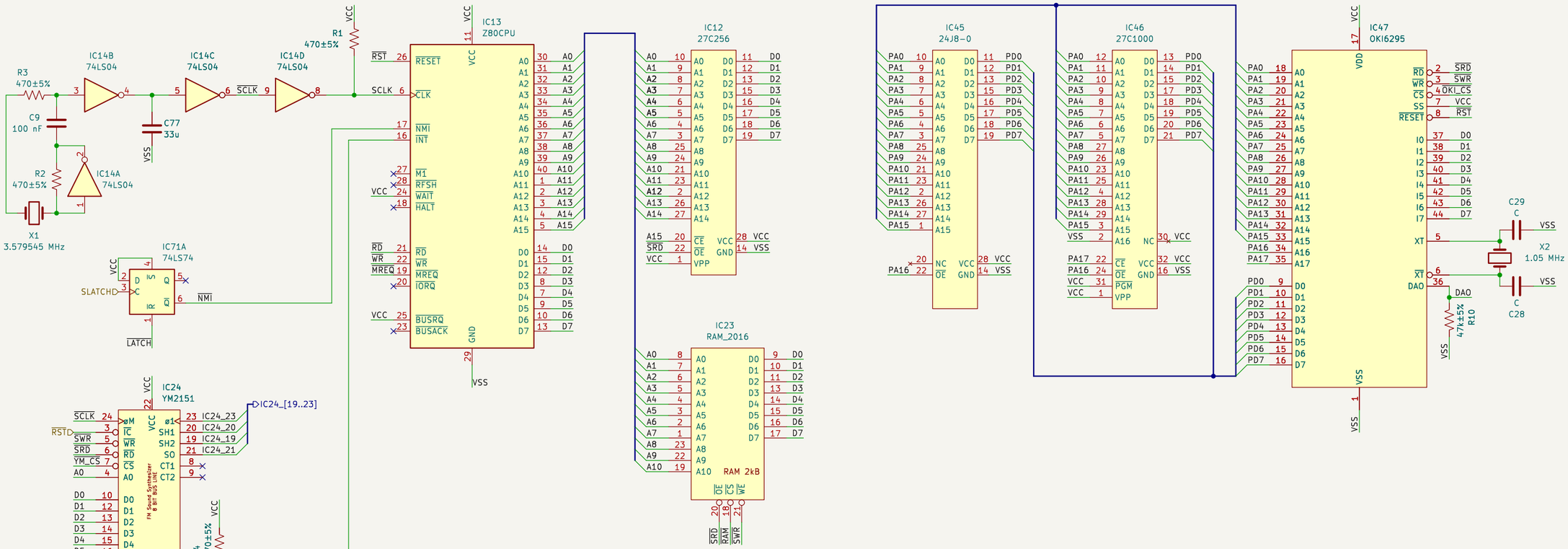








JOTEGO		
Sheet: /bottom/		
File: bottom.kicad_sch		
Title: Bottom board		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 3/20



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Sheet: /top/sound/
File: sound.kicad_sch

Title: WWF Super Stars

Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 4/20

MA[1..11]
MD[0..7]
CHA[0..11]
IC22
RAM_2016
IC32
buffers
74LS245
IC42
IC43
IC31
IC41
IC40
flip flops
74LS273
hex flip flop
74LS174
RAM 2kB
IC9
IC10
IC11
Quad Mux
74LS157
RAMOE
RAMWE
VSS
VCC
RWB
MCHAR
CHSEQ0
CHSEQ1
CHSEQ2
LDR0
LDR1
LDR2
LDS
AS
SCR
CHAR
PRECHDTACK
MA1
V8
IC14_13
IC82_12
IC14_11
IC94_11
IC44_1
IC44_2
IC58_8
IC58_9
IC58_10
IC58_11
IC58_12
PREPXL[0..3]
CHAPXL[0..6]
CHDTACK
char pixels
File: chpixel.kicad_sch
JOTEGO
Sheet: /top/char/
File: char.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.9
Id: 5/20

IC63
68000D
CLK
10 MHz
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
A17
A18
A19
A20
A21
A22
A23
VCC
VSS
GND

IC75
IC62
IC2
IC93
74LS244
74LS245
buffers
A->B
MA[1..16]
MD[0..15]

IC35
27C1000
IC34
27C1000
IC16
RAM_2016
IC15
RAM_2016
8kB
RAM

IC74
74LS74
IC66B
IC66A
IC65
74LS148
IC25C
74LS10
IC18A
74LS32
IC17C
74LS10
IC17B
74LS10
IC48
74LS138
decoder
IC50
74LS138
decoder
IC67
74LS174
hex flip flop

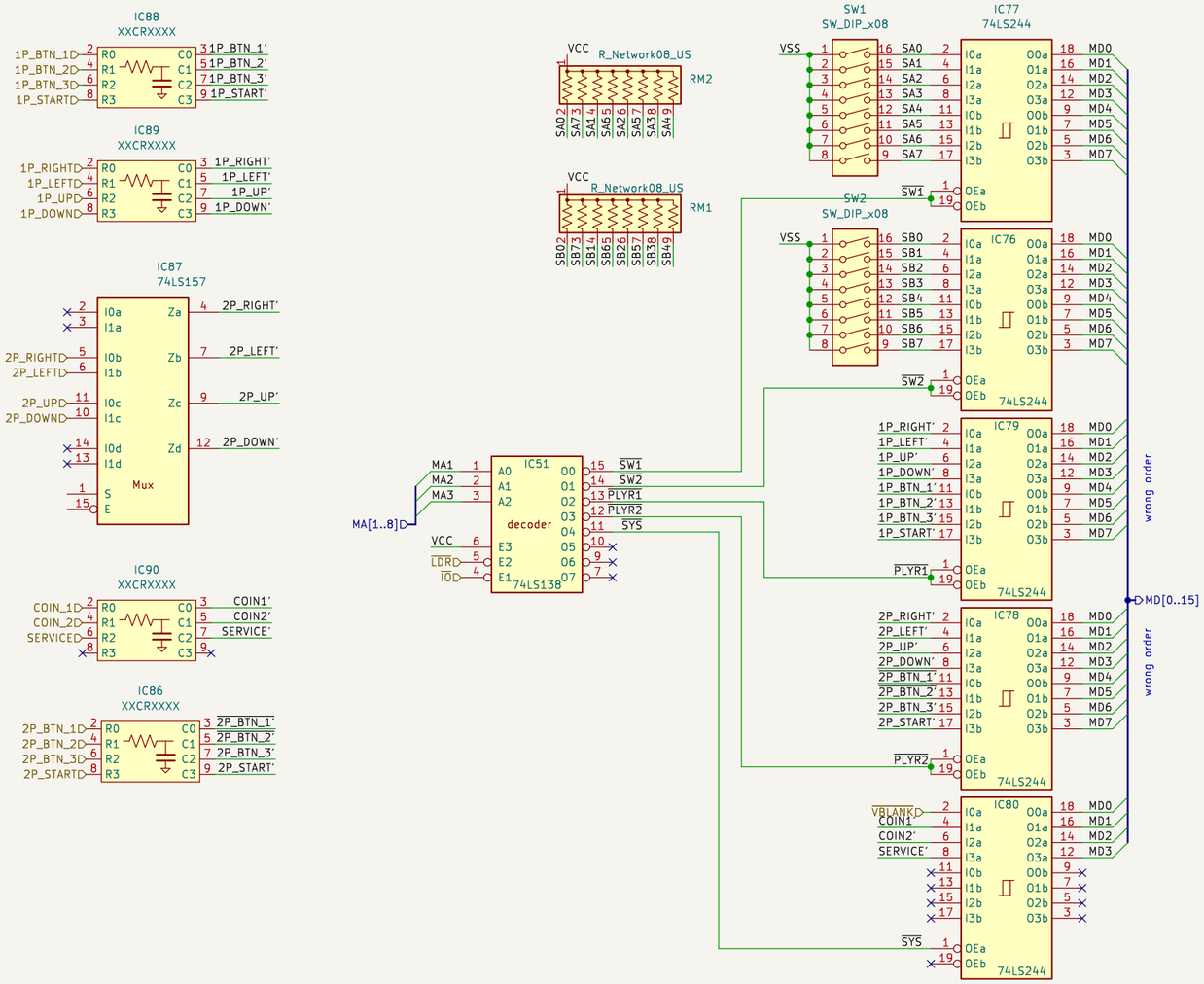
V8D
every
16 lines
IRQ5_CLR
VBLANK
IRQ6_CLR
ROM
LDR
UDR
CE
OE
PGM
VPP
WR
RAM
CS1
CS2
OEa
OEb
CHAR
SCR
OBJ
PAL
SCRX
SCRY
SLATCH
LOCK2
LOCK1
IC67_12
IC67_15

R23
47±5%
10k±5%
R22
47±5%
R13
47±1%
R12
47±1%
C62
C_Polarized_US
power-on reset
IC84A
74LS14
IC83B
74LS08
IC83A
74LS08
IC84B
74LS14
IC84C
74LS14
RST

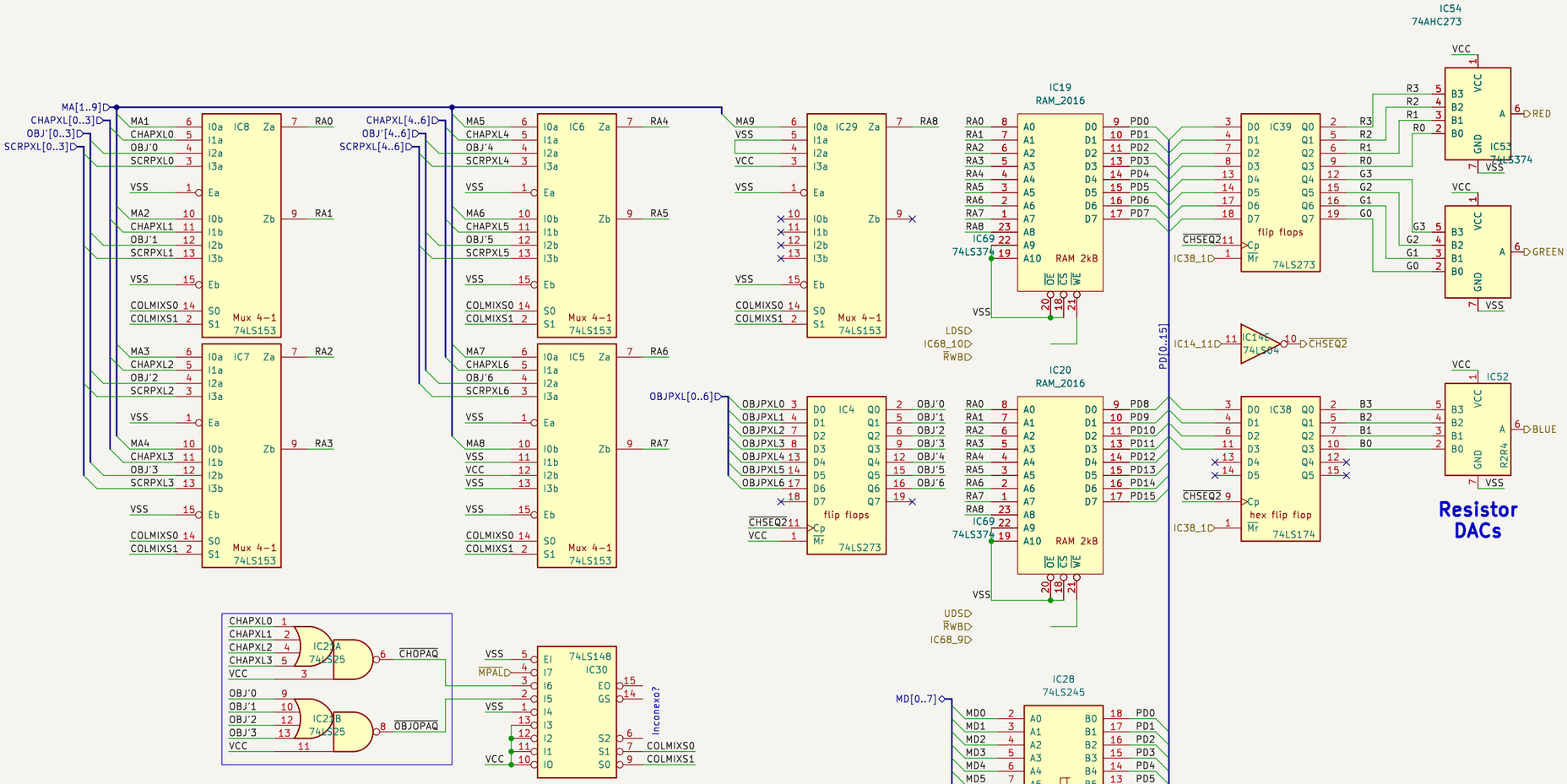
RFIX1
47±1%
33u
CFIX1
piggyback
soldered
X3
20.000MHz
IC82A
74LS74
CLK
25V
C61

IC49
IC44_1D
IC64H
IC64G
IC64E
IC64F
74LS04
IC1A
74LS00
IC64D
IC37D
IC37A
IC37C
IC36A
IC25B
IC59B
IC59C
IC3A
IC18D
IC72B
IC68C
IC68B
IC1D
IC17A
IC1B
IC1C
IC18B
IC83C
IC72D
IC72F
IC64G
IC1E
IC17D
IC66C
IC83G
IC21C
74LS74
74LS05
74LS25
IC59_6
IC25_5
IC68_10
IC68_9
IC28_CE
IC27_CE
RW_VID
MPAL
LDW
UDW
DTACK
CHDTACK

JOTEGO
Sheet: /top/main/
File: main.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.9
Id: 6/20



JOTEGO	
Sheet: /top/io/	
File: io.kicad_sch	
Title: CABINET I/O	
Size: A3	Date:
KiCad E.D.A. 8.0.9	Rev: Id: 7/20

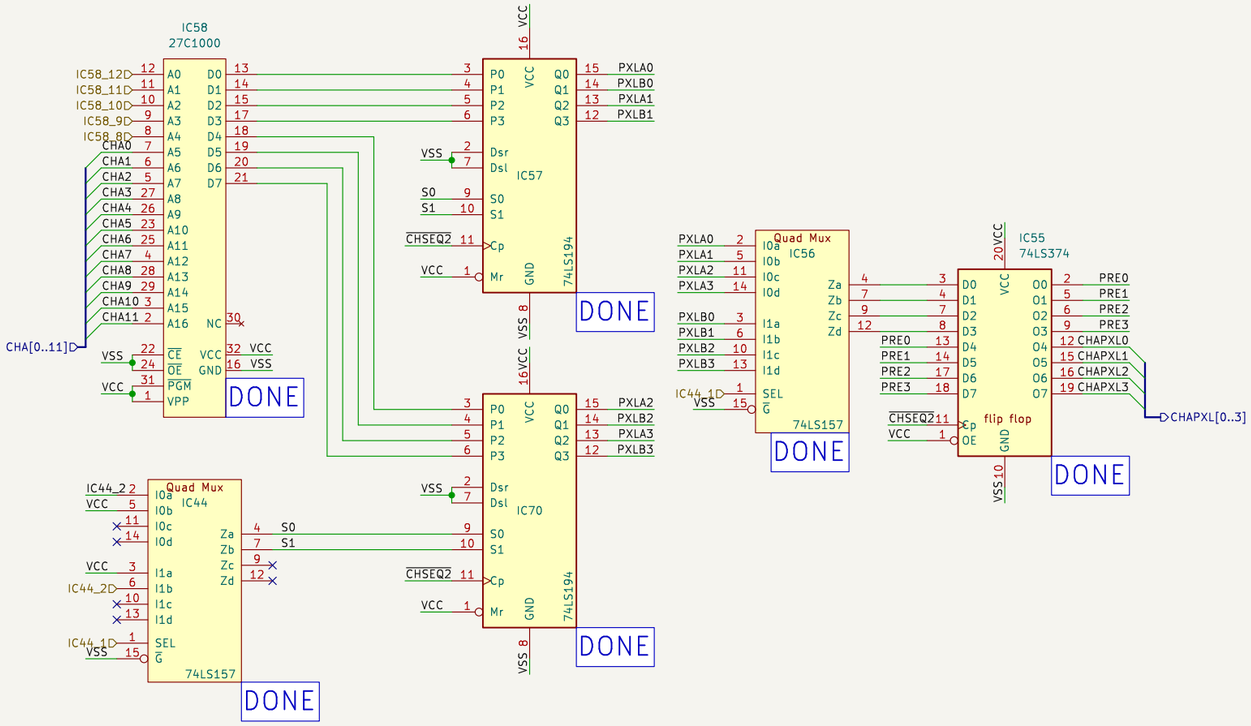


JOTEGO

Sheet: /top/color mixer/
File: colmix.kicad_sch

Title:

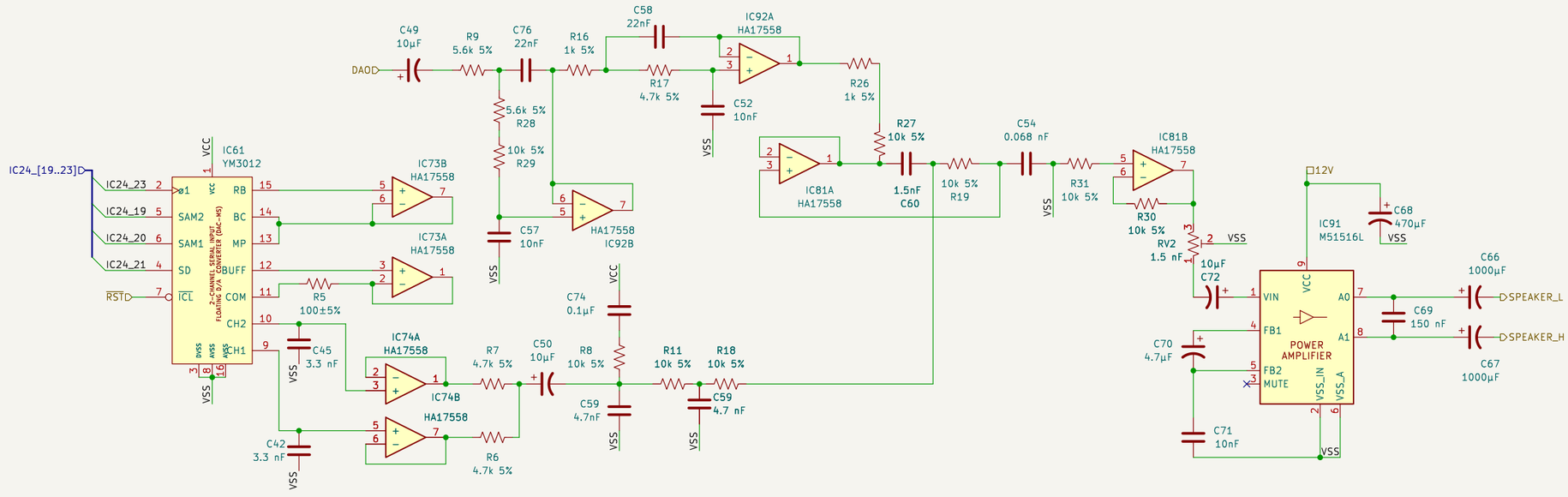
Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 8/20

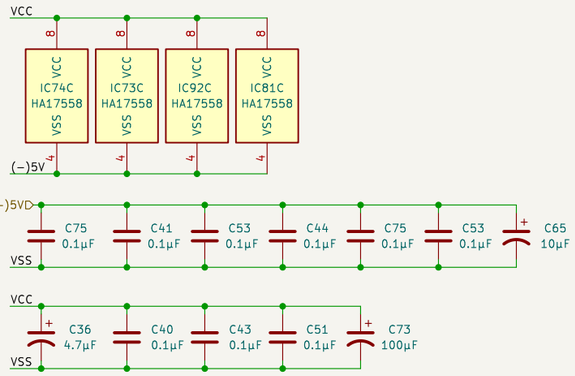


JOTEGO
Sheet: /top/char/char pixels/
File: chpixel.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.9		Id: 9/20



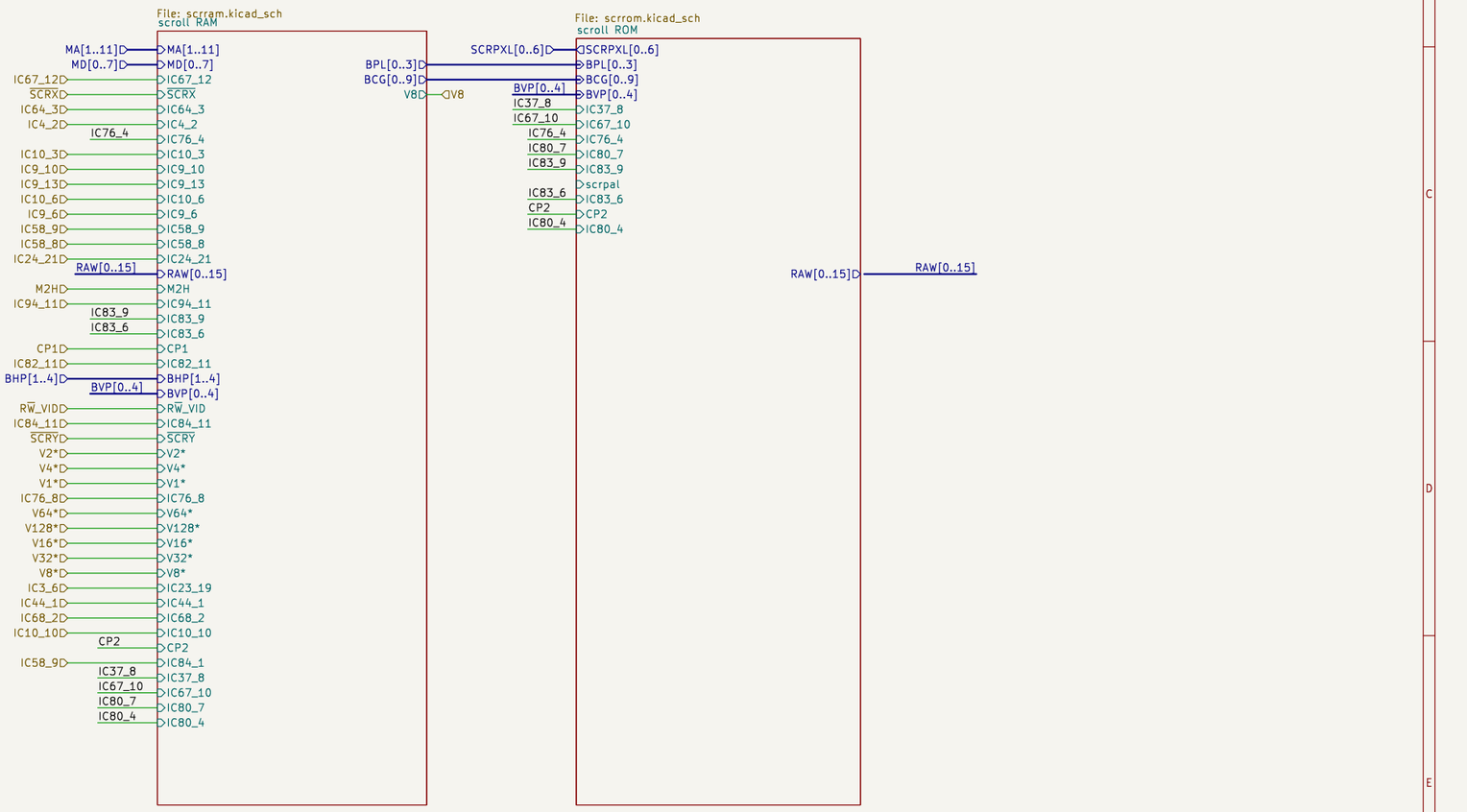


JOTEGO

Sheet: /top/sound/analog/
File: analog.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 10/20

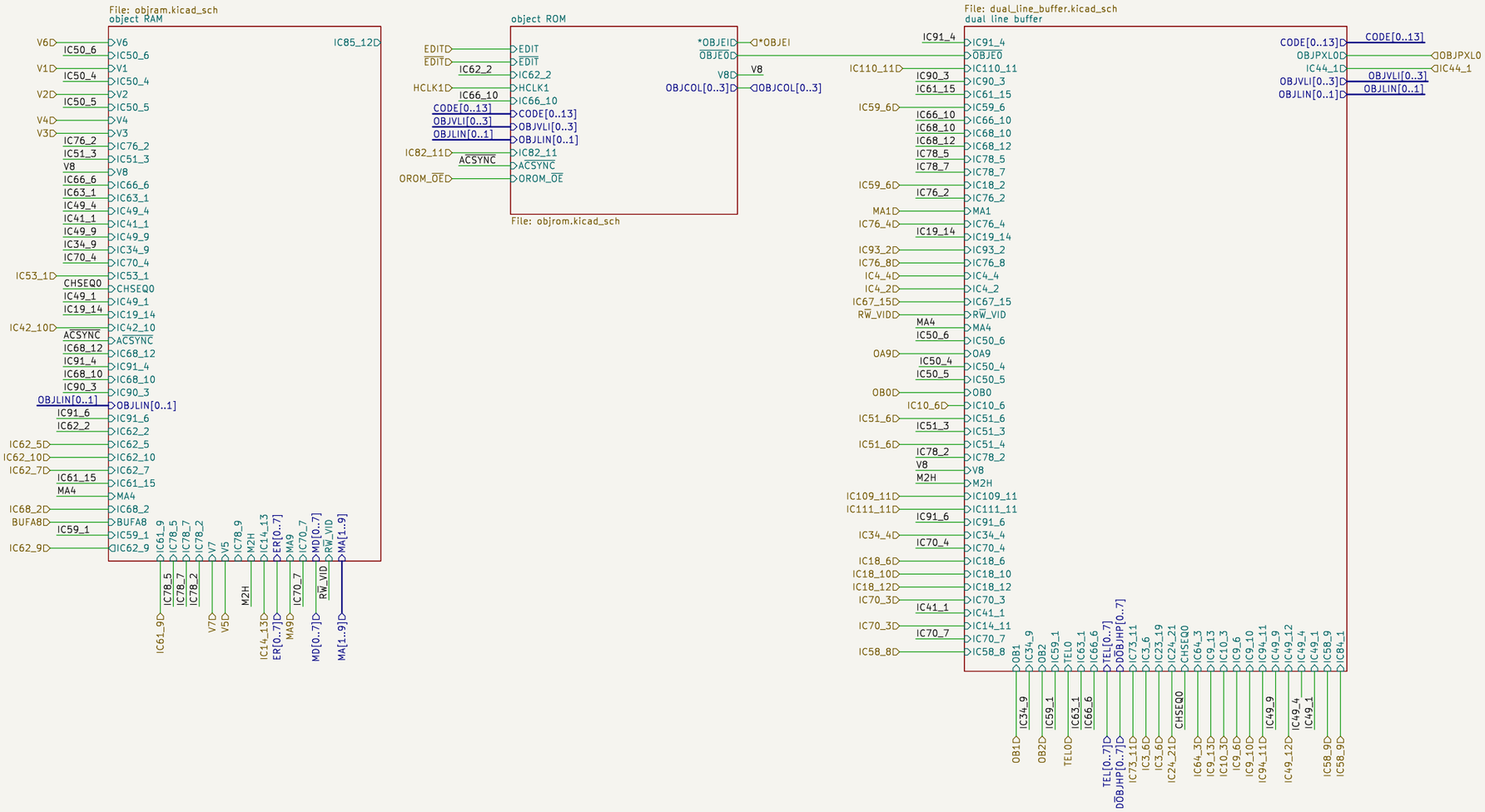


JOTEGO

Sheet: /bottom/scroll/
File: scroll.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 12/20



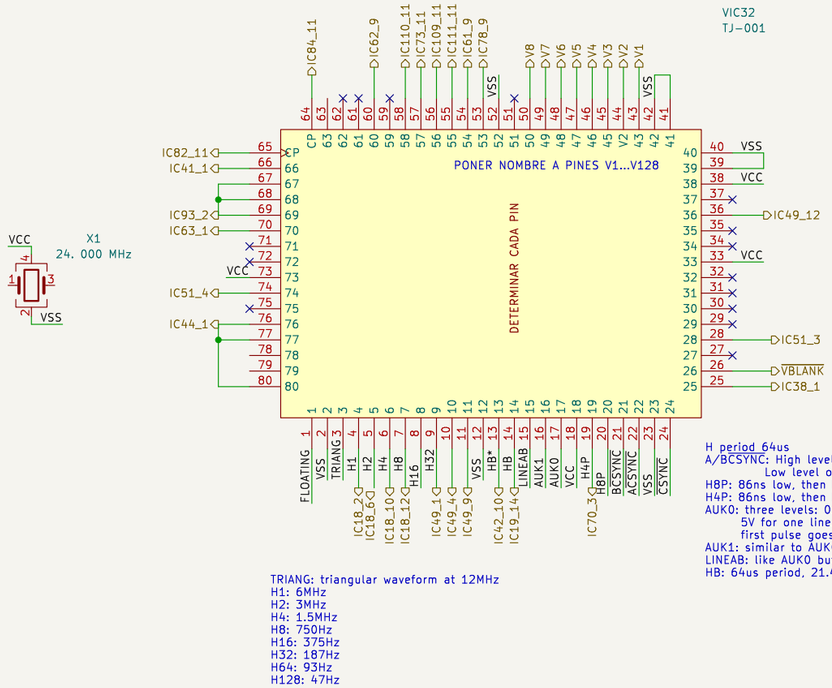
JOTEGO

Sheet: /bottom/object/

File: object.kicad_sch

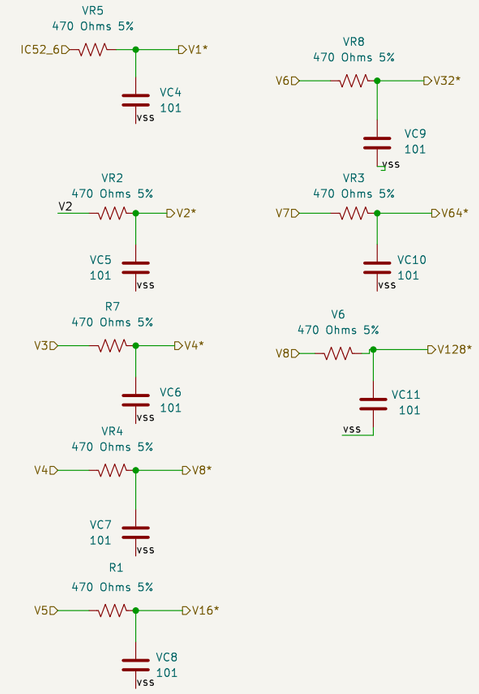
Title: Objects

Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.9		Id: 13/20

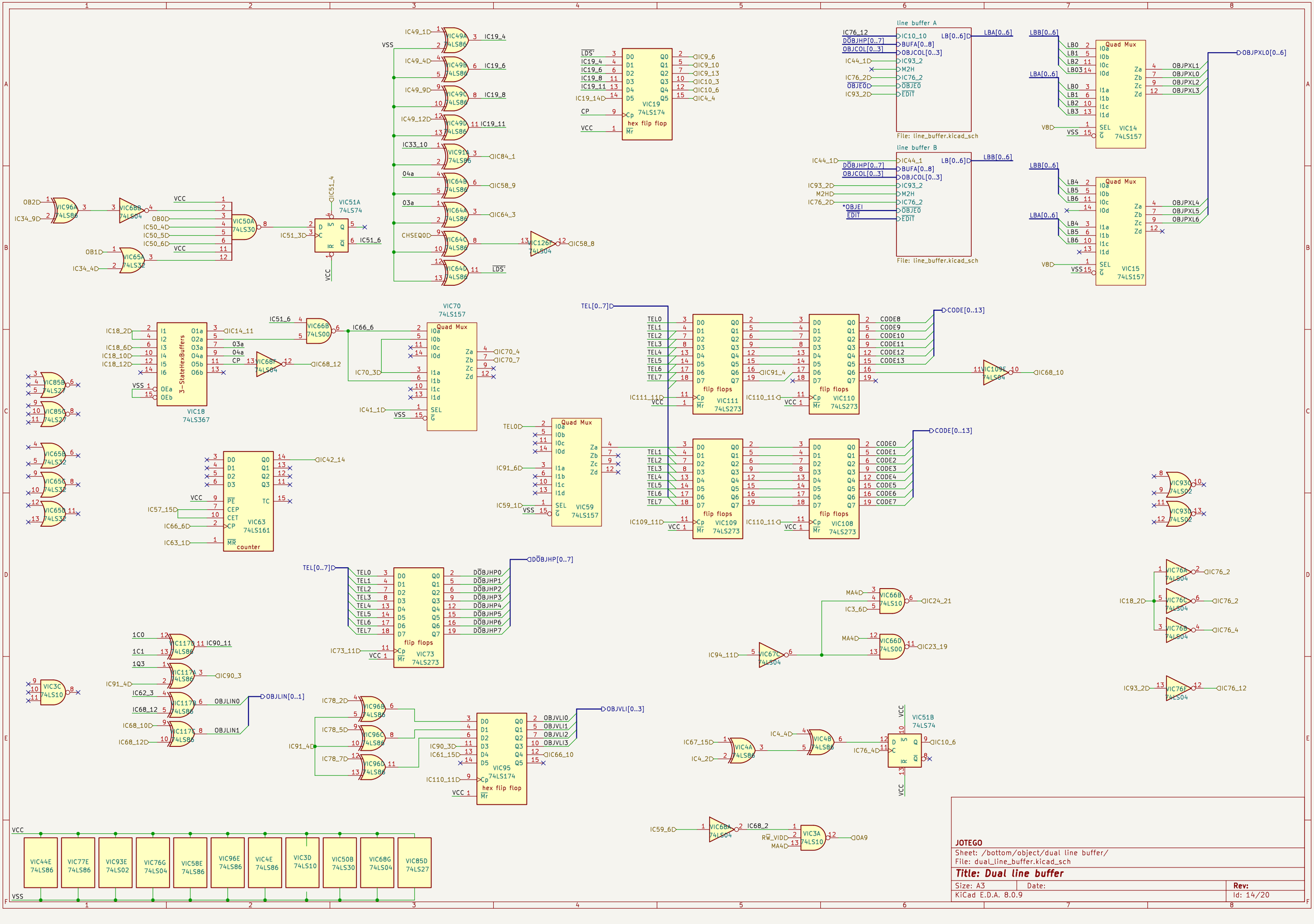


TRIANG: triangular waveform at 12MHz
H1: 6MHz
H2: 3MHz
H4: 1.5MHz
H8: 750Hz
H16: 375Hz
H32: 187Hz
H64: 93Hz
H128: 47Hz

H period 64us
A/BCSYNC: High level on visible area 5V, during VBLANK ~2V
Low level on visible area ~2V, during VBLANK 0V
H8P: 86ns low, then high. Period 668ns
H4P: 86ns low, then high. Period 332ns
AUK0: three levels: 0, 2 and 5V. subperiod 2.66us. period 128us
5V for one line, then pulses down during the next line. The first pulse goes 5->0V, the rest 5->2V
AUK1: similar to AUK0
LINEAB: like AUK0 but purely digital (0/5V)
HB: 64us period, 21.4us high

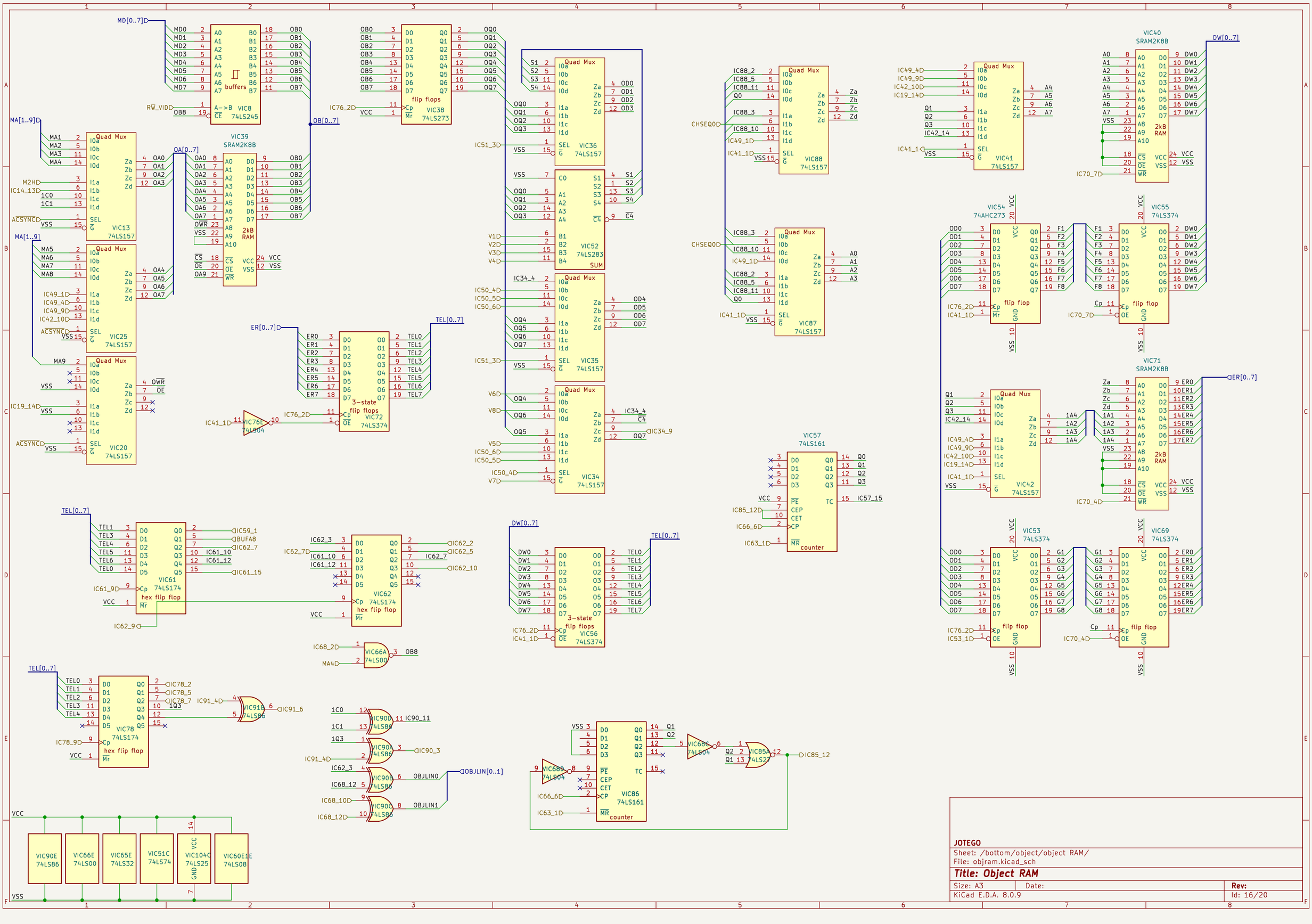


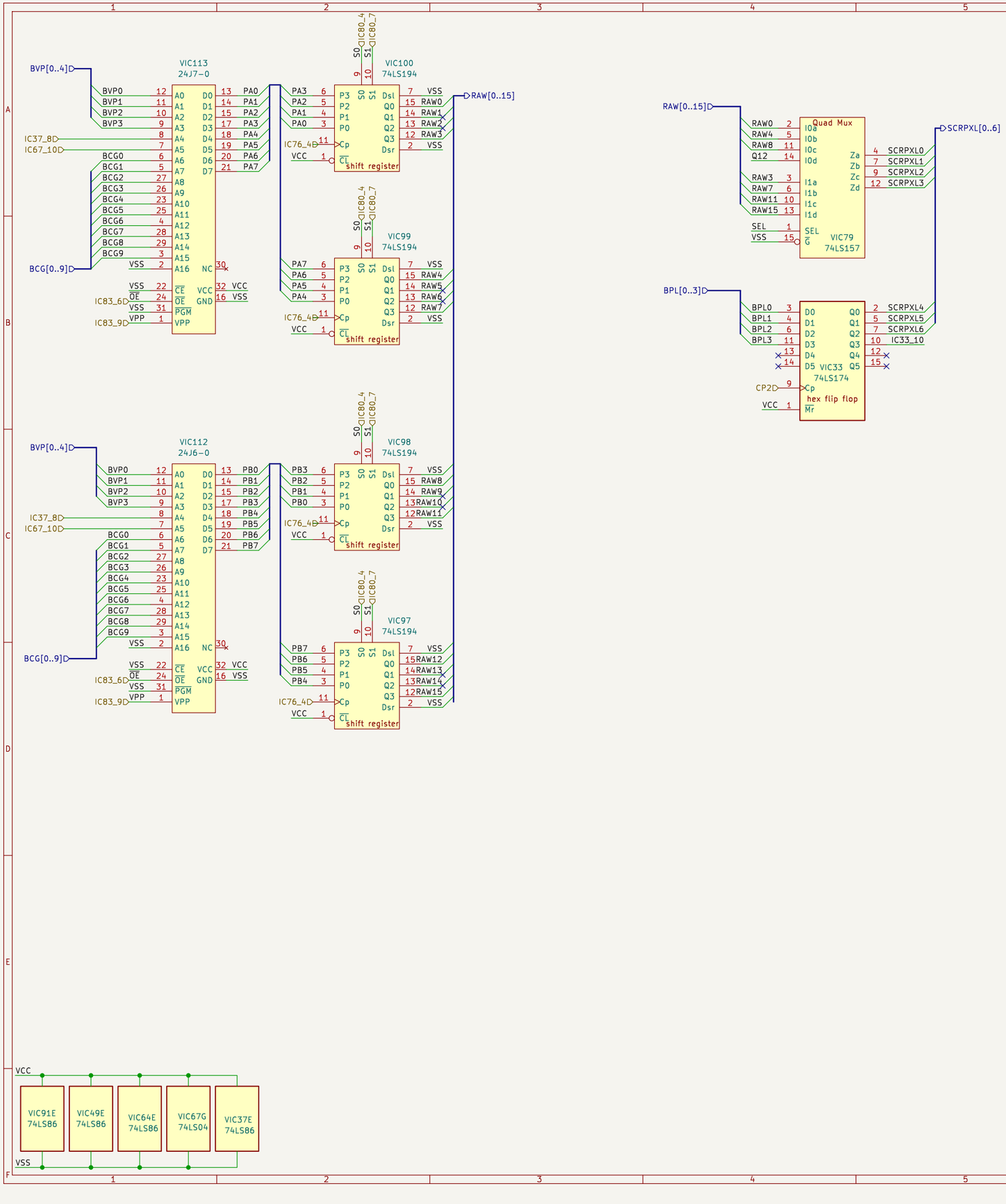
JOTEGO		
Sheet: /bottom/timing/		
File: timing.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 8.0.9		Id: 13/20


line buffer A
File: line_buffer.kicad_sch
line buffer B
File: line_buffer.kicad_sch
Quad Mux
flip flops
hex flip flop
3-StateHexBuffers
counter
JOTEGO
Sheet: /bottom/object/dual line buffer/
File: dual_line_buffer.kicad_sch
Title: Dual line buffer
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.9
Id: 14/20

VU117 TC534000AP
VIC114 TC534000AP
VIC115 24J 256 kB MASK ROM
VIC116 24J 256 kB MASK ROM
VIC118 24J 256 kB MASK ROM
VIC119 24J 256 kB MASK ROM
512kB MASK ROM
VIC103 74LS194
VIC102 74LS194
VIC107 74LS194
VIC106 74LS194
shift register
VIC104A 74LS25
VIC60A 74LS08
VIC60B 74LS08
VIC60D 74LS08
VIC105 74LS157 Quad Mux
VIC6BA 74LS04
VIC64B 74LS04
VIC66C 74LS00
VIC101 74LS158
OBJVLI[0..3]
OBJLIN[0..1]
CODE[0..13]
OBJCOL[0..3]
OBJEO
*OBJEI
EDIT
V8
ACSYNCD
HCLK1D
IC82_11D
IC66_10D
IC62_2D
-> CE
2 -> A16
ambiar posiciones de pines y cables

JOTEGO
Sheet: /bottom/object/object ROM/
File: objrom.kicad_sch
Title: Object ROM
Size: A3 | Date: | Rev:
KiCad E.D.A. 8.0.9 | Id: 15/20


MD[0..7]
MD0
MD1
MD2
MD3
MD4
MD5
MD6
MD7
RW_VID
OB8
A0
A1
A2
A3
A4
A5
A6
A7
A–>B
CE
buffers
VIC8
74LS245
B0
B1
B2
B3
B4
B5
B6
B7
OB0
OB1
OB2
OB3
OB4
OB5
OB6
OB7
OB[0..7]
IC76_2
VCC
D0
D1
D2
D3
D4
D5
D6
D7
Cp
Mr
flip flops
VIC38
74LS273
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
OQ0
OQ1
OQ2
OQ3
OQ4
OQ5
OQ6
OQ7
MA[1..9]
MA1
MA2
MA3
MA4
M2H
IC14_13
1C0
1C1
ACSYNC
VSS
Quad Mux
I0a
I0b
I0c
I0d
I1a
I1b
I1c
I1d
Za
Zb
Zc
Zd
SEL
G
VIC13
74LS157
OA[0..7]
OA0
OA1
OA2
OA3
OA4
OA5
OA6
OA7
VIC39
SRAM2K8B
OWR
OE
CS
WR
2kB
RAM
VCC
MA5
MA6
MA7
MA8
IC49_1D
IC49_4D
IC49_9D
IC42_10D
VIC25
74LS157
MA9
IC19_14D
VIC20
74LS157
ER[0..7]
ER0
ER1
ER2
ER3
ER4
ER5
ER6
ER7
3-state
flip flops
VIC72
74LS374
TEL[0..7]
TEL0
TEL1
TEL2
TEL3
TEL4
TEL5
TEL6
TEL7
IC41_1D
VIC76E
74LS04
IC76_2D
IC34_9
IC34_4
C4
S1
S2
S3
S4
C0
A1
A2
A3
A4
B1
B2
B3
B4
SUM
VIC52
74LS283
V1
V2
V3
V4
IC51_3D
VIC36
74LS157
IC50_4D
IC50_5D
IC50_6D
VIC35
74LS157
V6
V8
V5
V7
VIC34
74LS157
IC88_2
IC88_5
IC88_11
IC88_3
IC88_10
IC49_1D
IC41_1D
CHSEQ0D
VIC88
74LS157
IC88_2D
IC88_5D
IC88_11D
Q0
Q1
Q2
Q3
VIC87
74LS157
IC49_4D
IC49_9D
IC42_10D
IC19_14D
IC41_1D
VIC41
74LS157
A4
A5
A6
A7
VIC40
SRAM2K8B
DW[0..7]
DW0
DW1
DW2
DW3
DW4
DW5
DW6
DW7
VIC54
74AHC273
OD0
OD1
OD2
OD3
OD4
OD5
OD6
OD7
F1
F2
F3
F4
F5
F6
F7
F8
VIC55
74LS374
VIC57
74LS161
PE
TC
CEP
CET
CP
MR
counter
IC85_12D
IC66_6D
IC63_1D
IC57_15
VIC42
74LS157
VIC71
SRAM2K8B
VIC53
74LS374
VIC69
74LS374
VIC56
74LS374
VIC66A
74LS00
IC68_2D
MA4D
VIC78
74LS174
IC78_9D
IC78_2
IC78_5
IC78_7
1Q3
VIC91B
74LS86
IC91_4D
IC91_6
VIC90D
VIC90A
VIC90B
VIC90C
IC90_11
IC90_3
OBJLIN0
OBJLIN1
OBJLIN[0..1]
IC62_3D
IC62_7D
IC61_10D
IC61_12D
IC68_12D
IC68_10D
VIC61
74LS174
hex flip flop
IC61_9D
IC62_9D
IC59_1
BUFAB
IC62_7
IC61_10
IC61_12
IC61_15
VIC62
74LS174
IC62_2
IC62_5
IC62_10
VIC68B
74LS04
VIC85A
74LS27
IC85_12
VIC90E
74LS86
VIC66E
74LS00
VIC65E
74LS32
VIC51C
74LS74
VIC104C
74LS25
VIC60E1E
74LS08
JOTEGO
Sheet: /bottom/object/object RAM/
File: objram.kicad_sch
Title: Object RAM
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.9
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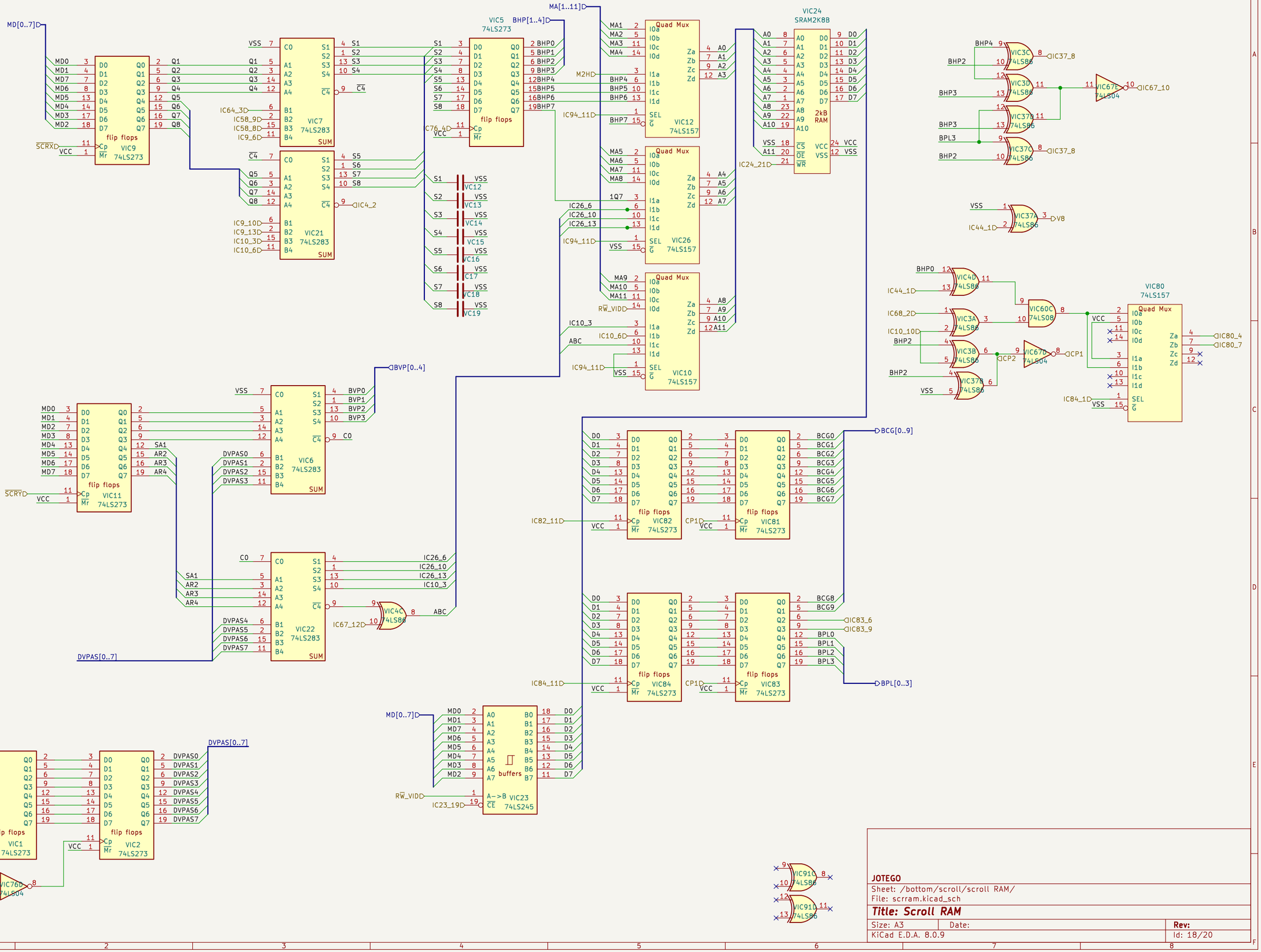
JOTEGO

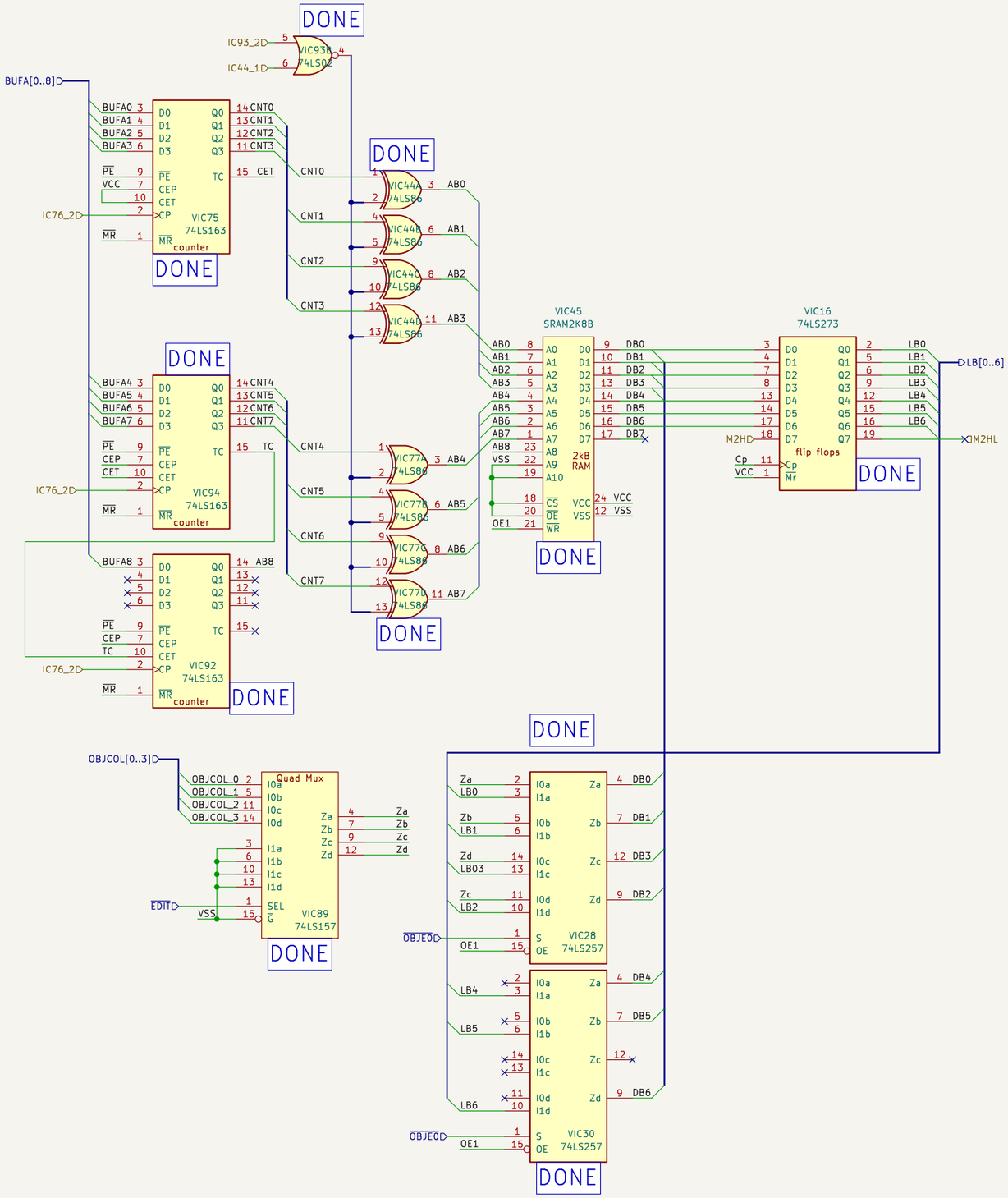
Sheet: /bottom/scroll/scroll ROM/

File: scrrom.kicad_sch

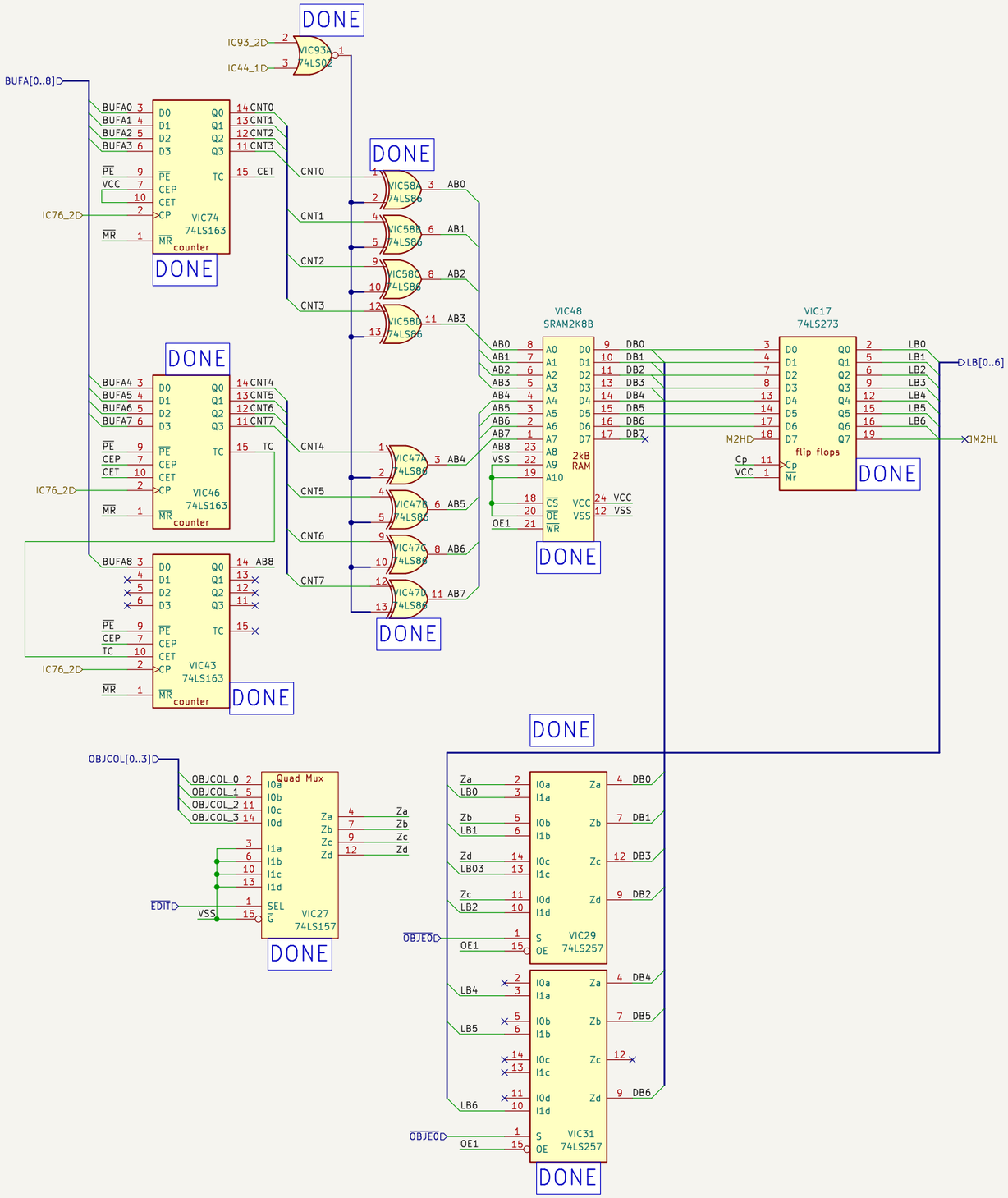
Title:

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MD[0..7]
MA[1..11]
BVP[0..4]
DVPAS[0..7]
BCG[0..9]
BPL[0..3]
VIC5 74LS273
VIC24 SRAM2K8B
VIC80 74LS157
VIC7 74LS283 SUM
VIC21 74LS283 SUM
VIC6 74LS283 SUM
VIC22 74LS283 SUM
VIC9 74LS273
VIC11 74LS273
VIC1 74LS273
VIC2 74LS273
VIC82 74LS273
VIC81 74LS273
VIC84 74LS273
VIC83 74LS273
VIC12 74LS157
VIC26 74LS157
VIC10 74LS157
VIC23 74LS245
flip flops
Quad Mux
buffers
2kB RAM
JOTEGO
Sheet: /bottom/scroll/scroll RAM/
File: scram.kicad_sch
Title: Scroll RAM
Size: A3
Date:
Rev:
KiCad E.D.A. 8.0.9
Id: 18/20



Sheet: /bottom/object/dual line buffer/line buffer A/		
File: line_buffer.kicad_sch		
Title: Line Buffer		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 20/20



Sheet: /bottom/object/dual line buffer/line buffer B/
File: line_buffer.kicad_sch

Title: Line Buffer

Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 21/20